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The Battle of Brains
An International Techno - Management Fest

K! HACKS 4.0

ONLINE SUBMISSION

Team Name : Type Here
College Name : Madras Institute of Technology
Problem Track : FinTech

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PROBLEM STATEMENT

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> *"...why your AI operated tools in the bank did not deem it fit to alarm him, that this transaction is suspicious?"*
>
> Chief Justice Surya Kant, Supreme Court hearing on digital-arrest scams, 9 Feb 2026 [2]

28 lakh
frauds reported on NCRP in 2025 (2.6 lakh in 2021) [1]

₹22,931 cr
reported lost in 2025 [1]

98.5%
of the money lost was in payments above ₹10,000 [1]

Scammers no longer steal passwords. They make the customer pay.

- **Right PIN, real phone, real customer.** RBI: most fraud is now authorised push payment; account takeover is "negligible" [1].
- **Victims are coached to lie.** Mulki couple: "buying land". Agra victim: "son's property". Both were scam payments [3][6].
- **RBI's proposed 1-hour lag** on every payment above ₹10,000 slows everyone, and victims can be told to whitelist the payee [1].

What saved the money

Navi Mumbai, Nov 2024 [5]

Manager saw the payee **"Royal enterprises"** did not fit the "medical" reason. **₹12 lakh saved**, after ₹21.94 lakh was gone.

Agra, Sep 2026 [3]

80-year-old closed his FDs to send ₹81 lakh. The bank **called his daughter-in-law**, who knew about the fake-police calls. **Transfer stopped.**

Hyderabad, Oct 2026 [4]

Six app transfers in five days, **₹1.5 crore**. Nobody stepped in.

In a branch, a person notices. In a mobile app, nobody does.

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SOLUTION ABSTRACT

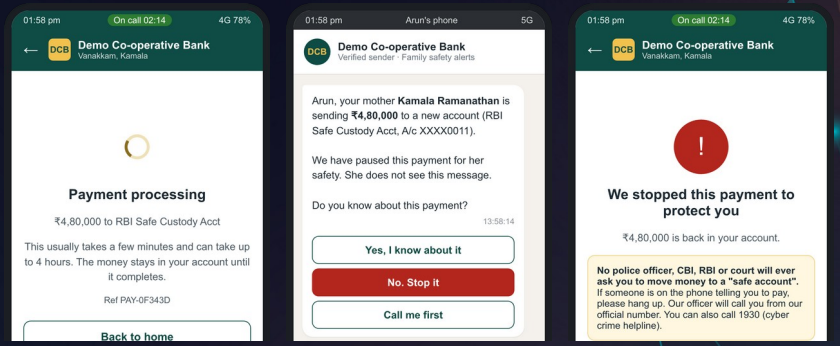
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SafePause: pause only the risky payments, without tipping off the scammer

Sits between the bank's app and its payment system. Scores every outgoing payment on **26 signals the bank and phone already have** (amount vs usual, new payee, FD just closed, call with an unknown number, pasted details, risk-listed receiver) in **about 25 ms**.

LOW	**Pays at once.** 99.5% of genuine payments in our test.
MEDIUM	**Silent pause, up to 30 min.** App shows "processing". The family member chosen at sign-up is asked: "Do you know about this payment?" Yes releases it at once.
HIGH	**Hold, up to 4 hours.** A bank officer sees the reasons in plain words, calls the customer, then stops it (money returned) or releases it.

Why not RBI's options as they are? Option 1 delays **every** payment above ₹10,000 by an hour. Option 2 needs a trusted person for **every** payment above ₹50,000 by anyone 70+. **SafePause uses the same two tools only when the risk is real.**



Kamala sees "processing" | Her son is asked | Stopped, money back

SIMULATED RESULT

83% of scam money stopped

0.48% of genuine payments delayed

1-hour lag: 11% stopped, 4.6% delayed

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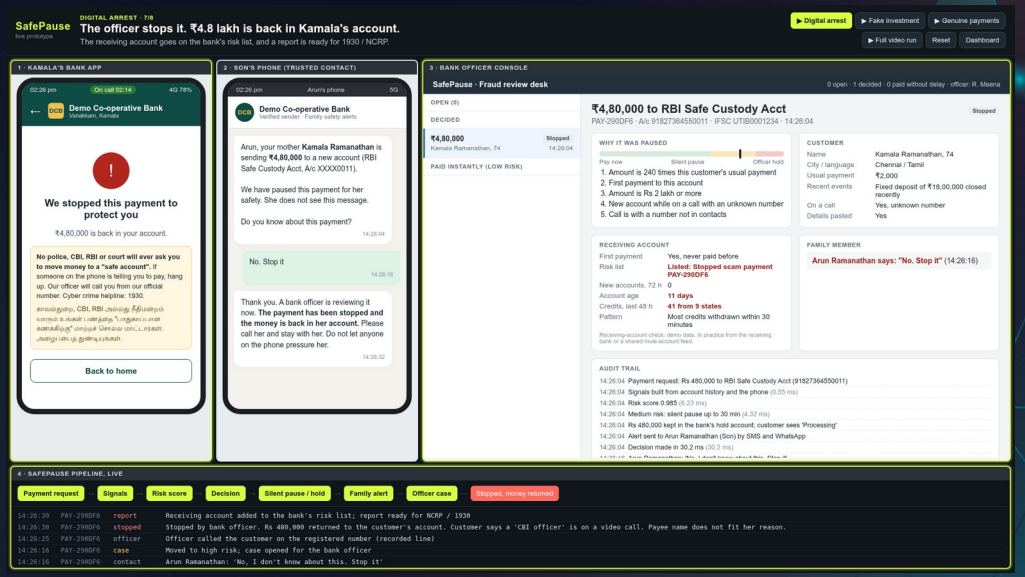
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ARCHITECTURE AND SYSTEM DESIGN

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INPUTS	SAFEPAUSE ENGINE	ACTIONS
Bank app SDK Call on? Number saved? Details pasted? Time to confirm. Screen sharing.	**1 Signals** 26 signals from the inputs	**Hold** Money stays in the bank; app says "processing"
Core banking Payment history, known payees, FD closures, credits from strangers.	**2 Risk score** Scorecard model with reason codes	**Family alert** SMS / WhatsApp: Yes / No / Call me
Risk feeds DoT FRI risky numbers, mule-account lists.	**3 Decision** Low / medium / high + safety rules	**Officer console** Reasons, payee checks, call script

Outcome release, or stop and return the money → audit log → NCRP / 1930 report → confirmed cases retrain the score



Live prototype → github.com/Sanmugavel1/SafePause

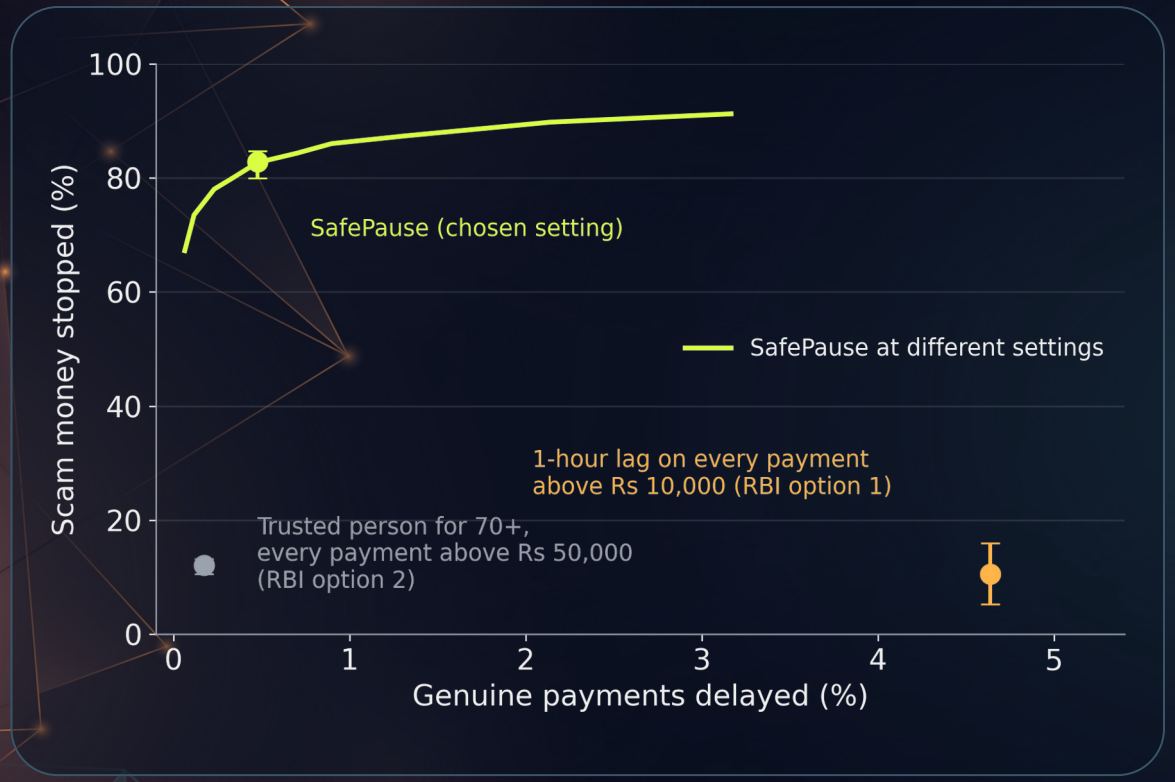
- **Working end to end:** Python + SQLite, REST API, 5 screens
- **17 to 30 ms** per decision, measured on a laptop
- **Same model** as the simulation on slide 5
- **Privacy:** no call audio or contacts read, only "call on, number not saved"

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FEASIBILITY AND UNIQUE SELLING POINTS

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Synthetic test month: 2.79 lakh genuine payments, 2,731 scam cases, 200 runs over unknown human behaviour. Bars: 90% range.

Why it stands out

- **10x fewer delays, 8x more money stopped** than a 1-hour lag on every payment.
- **Silence matters:** if the app openly says "held", the scammer moves the victim to another bank. Money stopped falls from 83% to 65%.
- **Catches scams it never saw:** task/job scams, left out of training: 90% stopped.

Real cases replayed

Case (news reports)	Lost	Paused at	No call signal
Hyderabad Oct 2026	₹1.50 cr	**payment 1 of 6**	payment 2
Jodhpur Feb 2026	₹2.07 cr	**payment 1 of 2**	payment 1
Navi Mumbai Nov 2024	₹33.9 lakh	**payment 1 of 4**	payment 3

Already done elsewhere: UK banks may hold a suspicious payment up to 4 business days (since Oct 2024) [7]. Google warns Indian users who open finance apps while screen sharing on unknown calls [8].

Weak spot: one-shot scams (fake KYC call, cloned-voice relative) only 39% / 35% stopped. Synthetic data: shadow pilot first.

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MARKET STUDY AND BUSINESS MODEL

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Who pays, and why now

- **Banks carry the loss.** RBI's revised fraud-liability rules apply from **1 Jan 2027** [10]. The Supreme Court wants a shared-liability and compensation framework [2].
- **Pausing everyone costs goodwill.** SafePause pauses about **1 in 200** genuine payments.

First customers

1,457	28	11
urban co-operative banks [11]	regional rural banks	small finance banks

Older, rural customers; small fraud teams. Next: UPI apps, then big banks via core-banking vendors.

Gap: DoT FRI scores phone numbers; MuleHunter.AI finds mule accounts on the receiving side; BioCatch, Feedzai, FICO are platforms for big banks. **SafePause acts on the sending side, at the moment of payment, with a family and officer loop.**

How we charge

proposed; to be tested in the pilot

₹3 lakh one-time setup per bank

₹2 per mobile-banking user per month

Partners core-banking vendors bundle it; cyber-fraud insurers co-pay

Example: a bank with 1 lakh mobile users

Scam losses its customers face	**≈ ₹4.1 cr / year**	(₹413 per UPI user [1][9])
SafePause fee	**₹24 lakh / year**	(about 6% of that)
Officer time	**≈ 22 reviews / day**	(one officer)

Path

Shadow pilot with one co-operative bank (score only, nobody paused), 3 months → compare with its fraud reports → **RBI Regulatory Sandbox** → paid roll-out

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RESEARCH AND REFERENCES

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How we tested it

- **Synthetic bank:** 30,000 customers, 60 days, 5.76 lakh payments, 6,000 scam cases of 6 types, built from published cases.
- **Calibrated to RBI:** payments ≥ ₹10,000 are 45% of scam payments by count and 98% by value (RBI: 45%, 98.5%) [1].
- **Fair test:** trained on month 1 with 4 scam types, tested on month 2 with all 6. AUC 0.98 seen, 0.97 unseen. Hard genuine cases kept in.
- **Unknown behaviour:** 200 runs with ranges for family replies, officer judgement and ignored warnings.

Voices from the field (published)

"Banks must realise they are trustees of public money." CJI Surya Kant, 9 Feb 2026 [2]

Fraudsters had "digitally arrested the elderly couple for many days." Mangaluru Police Commissioner, Dec 2025 [6]

In-call warnings "helped thousands of users end calls that could have cost them a significant amount of money." Google, UK pilot, Dec 2025 [8]

Our interviews: [add 2-3 quotes from bank staff and older customers]

References

[1] RBI, Discussion Paper: Exploring safeguards in digital payments to curb frauds, 9 Apr 2026
[2] Supreme Court: hearing of 9 Feb 2026 (LiveLaw); order of 4 Aug 2026, 2026 SCC OnLine SC 1532
[3] Agra, Canara Bank case, 8 Sep 2026 (news report, 10 Sep 2026)
[4] The420.in, Hyderabad digital arrest fraud, Oct 2026
[5] Free Press Journal, Navi Mumbai bank manager saves 80-year-old, 29 Nov 2024
[6] ETV Bharat, Mangaluru manager saves couple from losing Rs 84 lakh, 5 Dec 2025
[7] UK Payment Services (Amendment) Regulations 2024; FCA guidance FG24/6
[8] Google Security blog, in-call scam protection for financial apps, 3 Dec 2025
[9] Lok Sabha reply: 55.49 crore UPI users, June 2026
[10] RBI, revised directions on fraudulent e-banking transactions, 24 Jun 2026 (from 1 Jan 2027)
[11] RBI, co-operative banks publication, 29 Dec 2025; FREE-AI report, 13 Aug 2025
[12] ETV Bharat, Jodhpur digital arrest case, 24 Feb 2026 (used in replay)

Code, simulation and live demo: github.com/Sanmugavel1/SafePause

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TEAM DETAILS

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ROLE	NAME	STREAM / DEPT	YEAR	COLLEGE
Team Lead				
Member				

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